# Prashant Gupta

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## Summary

Data Science and Machine Learning professional with hands-on experience building NLP pipelines, multi-agent AI systems, and predictive ML models from raw data to production deployment. Skilled in Python, Scikit-learn, LangChain, and REST API development, with measurable improvements in accuracy, efficiency, and system reliability across full-stack AI applications.

## Education

B.Tech at **Madan Mohan Malaviya University of Technology** — *2023 – Present*
Electrical Engineering — CGPA: 8.39/10
Minor Degree – Data Science — CGPA: 9.0/10

## Experience

**Founding Engineer – 2 | OBLIQ-in** — *Sept 2026 – Present*

- Building the AI/ML layer for an audit intelligence platform, analyzing journal-entry data to identify explainable and review-worthy transaction patterns.
- Exploring ML-based anomaly detection to complement deterministic risk rules and surface unusual account, amount, timing, and transaction patterns.
- Contributing to AI/OCR document intelligence for extracting structured information from invoices and audit evidence within a human-in-the-loop workflow.

**Research Intern | Nano-Intelligence Lab | IIT Roorkee** — *May 2026 – June 2026*

- Designed and developed an Edge AI-enabled Smart Glove using ESP32 and 4 flex sensors to recognize 5 hand gestures and convert them into real-time messages transmitted wirelessly via WiFi.
- Collected, labeled, and curated a gesture dataset of 1,586 high-quality samples across 5 balanced gesture classes with zero missing or corrupted values; achieved a signal-to-noise ratio (SNR) of ∼21
- Trained a TensorFlow Lite Micro model for TinyML deployment on ESP32, achieving 94% gesture classification accuracy with on-device edge inference and zero cloud dependency.

## Projects

**AI Business Operations Copilot – Multi-Agent RAG System** — Live Demo
*Python, FastAPI, Streamlit, Llama 3, Ollama, NumPy vector search, Sentence-Transformers*

- Designed a 4-agent sequential RAG pipeline (Planner → Researcher → Executor → Reviewer) using Sentence-Transformers (all-MiniLM-L6-v2) for semantic embeddings and L2 vector similarity search to retrieve and answer complex business queries with confidence scoring.
- Developed a full-stack application with a 6-page Streamlit UI and FastAPI backend, featuring document ingestion, configurable top-k retrieval, per-agent latency tracking, and a session-level evaluation dashboard.

**STATEMENTIQ — Bank Statement Intelligence** — Live Demo
*Python, Streamlit, pdfplumber, PyMuPDF, OpenCV, Tesseract, scikit-learn, pandas*

- Built a local, LLM-free pipeline that turns bank statement PDFs and images into validated, categorized transactions, using pdfplumber for text PDFs and OpenCV + Tesseract OCR for scans, with CSV and Excel export.
- Designed a rules + ML hybrid classifier (TF-IDF + Logistic Regression) and a running-balance validator that flags uncertain rows for review, backed by 250 pytest tests.

## Technical Skills

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| **Programming Languages** | Python, SQL |
| **ML & Data Science** | Machine Learning, Deep Learning, NLP, EDA, Feature Engineering |
| **Frameworks & Libraries** | Scikit-Learn, Pandas, NumPy, Matplotlib, Plotly, LangChain, LangGraph |
| **Generative AI** | LLM, RAG, Agentic AI, Prompt Engineering, Fine Tuning, Vector Database |
| **Cloud/Database/API** | Streamlit, MySQL, Flask, REST APIs, OpenAI API |
| **Mathematical Foundations** | Statistics, Linear Algebra, Probability, Algorithms |
| **Developer Tools/Version Control** | Codex, Claude Code, Antigravity, Cursor, GitHub Copilot, Git, GitHub |

## Certifications

Oracle Cloud Infrastructure 2025 Certified Data Science Professional — *Oct 2025*
Oracle Cloud Infrastructure 2025 Certified AI Foundation Associate — *Oct 2025*